

G-CAF

GovDigital Composable Architecture Framework

Volume 8

Templates & Catalogs

v2.0

Table of Contents

Chapter 1 Enterprise Catalogs Overview

The Catalog Quad and governance framework

Chapter 2 PBC Catalog

Authoritative registry of all Packaged Business Capabilities

Chapter 3 Technology Standards Catalog

Strategic/Tactical/Containment/Retired lifecycle

Chapter 4 Data Asset Catalog

Data entities, ownership, quality, and lineage

Chapter 5 Integration Asset Catalog

APIs, Events, Schemas with consumer contracts

Chapter 6 Technical Debt Management

Debt taxonomy, scoring, and remediation

Chapter 7 Reference Architecture Patterns

40+ approved reusable patterns across 6 categories

Chapter 8 Environment, Accessibility & Skills Standards

DEV→PROD environments, WCAG, skills matrix

Chapter 9 Vendor RFI Templates

Complete procurement template suite

Chapter 10 Golden Paths & Implementation Templates

PBC starter templates and standard paths

Chapter 1

Enterprise Catalogs Overview

G-CAF governance relies on a set of enterprise-wide catalogs that track capabilities, technologies, data, integrations, and more. All catalogs implement the Catalog Extension Framework (v1.18) which provides a unified metamodel, lifecycle management, quality governance, and cross-catalog linking.

The Catalog Quad

Catalog	Module ID	Purpose	Version
PBC Catalog	G-CAF-042	WHAT we build (Capabilities)	v1.0
Technology Standards Catalog	G-CAF-043	HOW we build (Technology)	v1.5
Data Asset Catalog	G-CAF-044	WHAT data (Information)	v1.6
Integration Asset Catalog	G-CAF-045	HOW they connect (APIs, Events)	v1.7

Supporting Catalogs

Catalog	Module ID	Purpose	Version
Technical Debt Management	G-CAF-046	Debt tracking and remediation	v1.8
Reference Architecture Patterns	G-CAF-047	40+ approved reusable patterns	v1.9
Environment Standards	G-CAF-048	DEV→SIT→UAT→STAGING→PROD specs	v1.10
Accessibility & Localization	G-CAF-049	WCAG compliance, RTL/Arabic	v1.11
Skills Matrix Standards	G-CAF-050	Role competencies and proficiency	v1.12
Governance Fact Model	G-CAF-051	Governance entity relationships	v1.0

Chapter 2

PBC Catalog

The PBC Catalog is the **authoritative registry** of all Packaged Business Capabilities across Abu Dhabi Government.

Catalog Fields

Field	Description
PBC ID / Name / Description	Unique identifier, name, and what the PBC does
Bounded Context	BC-XXX semantic boundary (Capability Tier)
Domain	Organizational ownership (GovData, GovOps, etc.)
PBC Type	Application, Analytics, Data, Orchestration, Digital Twin, Agent
Criticality	Critical, High, Medium, Low
Sourcing	Build, Buy, Outsource; actual sourcing details
Stability Tier	Foundational, Stable, Evolving
Status	Proposed, In Development, Active, Deprecated, Retired
Owner / Version	Owning entity/team; current internal version
Interface Contract Version	Current stable interface version
Standard Contract Versions	Declared compliance versions
APIs / Dependencies / Consumers	Published APIs, PBC dependencies, consuming services
Gateway Status	Last gateway passed
Alternate Supplier	Pre-qualified alternate vendor (Critical/High Buy PBCs)

Catalog Rules

Rule	Enforcement
No PBC deployed without Catalog registration	Go-Live Gateway requirement
Reuse assessment before creating new PBC	Initiation Gateway requirement
Deprecated PBCs must have migration path	DDA approval for deprecation
Retired PBCs must have zero consumers	Retirement Gateway requirement
All PBCs must declare standard contract versions	Initiation Gateway requirement

Rule	Enforcement
Interface changes require consumer notification	DDA-managed process

Chapter 3

Technology Standards Catalog

Strategic governance over technology choices across all PBCs, defining approved technologies by category and lifecycle status.

Technology Lifecycle Status

Status	Description	Usage	Governance
Strategic	Preferred, invest in adoption	Default for new PBCs	Automatic approval
Tactical	Allowed with justification	Approved by DDA on case basis	DDA approval required
Containment	Legacy, sunset planned	Migration required	Migration plan mandatory
Retired	Prohibited, no use	Cannot be used	ARB exception only

Technology Categories

Tier	Categories
Data Tier	Relational DB, NoSQL DB, Cache, Search Engine, Data Warehouse
Application Tier	Backend Runtime, Frontend Framework, Mobile Platform, API Framework, Testing
Integration Tier	API Gateway, Event Broker, Service Mesh, ETL/ELT, iPaaS
Infrastructure Tier	Cloud Provider, Container Platform, Serverless, IaC Tools
Security Tier	Identity (IAM), Secrets Management, WAF/DDoS, SIEM, Vulnerability
Operations Tier	Monitoring, Logging, Tracing, CI/CD, GitOps

Chapter 4

Data Asset Catalog

Enterprise-wide governance for data entities — authoritative definitions, ownership, quality standards, and lifecycle management.

Catalog Sections by Change Driver

Section	Change Driver	Update Trigger
Data Domain Classification	Organizational restructuring	Domain boundaries change
Data Entity Registry	Entity definition	New entity or attribute change
Master Data Governance	Ownership policy	SoT assignment rules change
Data Quality Standards	Quality requirements	Quality thresholds change
Data Lifecycle Governance	Lifecycle policy	Retention rules change

The Data Asset Catalog integrates with the PBC Catalog (WHAT we build), Technology Catalog (HOW we build), and Integration Catalog (HOW they connect) to form the complete Catalog Quad.

Chapter 5

Integration Asset Catalog

Enterprise-wide governance for APIs, Events, and Schemas — definitions, versioning, lifecycle, and discoverability.

Catalog Scope

In Scope	Out of Scope
API definitions and contracts	API implementation code
Event types and schemas	Event handler implementation
Schema registry governance	Data transformation logic
API/Event versioning and lifecycle	Infrastructure configuration
Consumer/Provider relationships	Performance tuning

Integration Module Relationships

Module	Integration Point
PBC Catalog (G-CAF-042)	PBCs own/consume APIs and Events
Data Asset Catalog (G-CAF-044)	APIs expose entities; Events propagate changes
Interface Contract (G-CAF-100)	Technical standards for interfaces
Integration Contract (G-CAF-072)	Integration patterns and ACL rules

Chapter 6

Technical Debt Management

Systematic identification, tracking, assessment, prioritization, and remediation of technical debt across all PBCs.

Debt Categories

Category	Examples	Impact
Code Debt	Duplicated code, complex functions, missing tests	Maintenance cost, bug risk
Architecture Debt	Monolithic design, tight coupling, missing capabilities	Agility loss, scaling issues
Infrastructure Debt	Manual deployments, outdated OS, legacy hardware	Reliability risk, security gaps
Documentation Debt	Missing API docs, outdated diagrams, no runbooks	Onboarding friction, knowledge loss
Testing Debt	Low coverage, no integration tests, manual QA	Quality risk, slow releases
Security Debt	Unpatched CVEs, weak auth, missing encryption	Breach risk, compliance failure
Data Debt	Inconsistent schemas, missing lineage, quality issues	Analytics failures, AI readiness

Technical debt affects all four catalogs: capability debt (PBC Catalog), technology constraints (Technology Catalog), data quality issues (Data Asset Catalog), and integration fragility (Integration Catalog).

Chapter 7

Reference Architecture Patterns

Pre-approved, reusable architecture patterns that PBCs can adopt. Each pattern describes the problem, context, forces, solution, consequences, and implementation guidance.

Pattern Categories

Category	Pattern Examples	Use Case
Integration Patterns	API Gateway, Event-Driven, Saga, CQRS, BFFs	Cross-PBC communication and data flow
Data Patterns	Medallion, Data Mesh, Data Virtualization, DQC	Data management and analytics
Security Patterns	Zero Trust, Defense in Depth, Fail Secure	Security architecture
Resilience Patterns	Circuit Breaker, Retry, Bulkhead, Fallback	Fault tolerance and availability
Deployment Patterns	Blue-Green, Canary, Feature Flags, GitOps	Safe production deployments
Composition Patterns	Micro-Frontend, BFF, Widget Registry, Shell	UI composition and assembly

Chapter 8

Environment, Accessibility & Skills Standards

Environment Standards (G-CAF-048)

Environment	Purpose	Data	Access
DEV	Development and unit testing	Synthetic data only	Developers
SIT	System integration testing	Sanitized production subset	Dev + QA
UAT	User acceptance testing	Production-like data	Business users + QA
STAGING	Pre-production validation	Production mirror	Operations + Release
PROD	Live production	Real data	End users

Accessibility & Localization (G-CAF-049)

Standard	Requirement
WCAG 2.1 AA	All public-facing experiences must comply
RTL/Arabic Support	Full right-to-left layout with Arabic typography
Bilingual Content	Arabic primary, English secondary; mirror layouts
Screen Reader	All interactive elements accessible via screen reader
Keyboard Navigation	Complete keyboard accessibility for all functions
Color Contrast	Minimum 4.5:1 ratio for text; 3:1 for large text

Skills Matrix Standards (G-CAF-050)

Defines role competencies and proficiency levels for all G-CAF roles: Entity Architect, Solution Architect, PBC Owner, Developer, Security Engineer, Data Engineer, Platform Engineer, and more. Each role has defined skills across Architecture, Development, Governance, and Operations domains with 4 proficiency levels (Awareness, Working, Practitioner, Expert).

Chapter 9

Vendor RFI Templates

Complete procurement template suite aligned with G-CAF and DGE IT Technical Requirements.

Template Suite

Template	Purpose	Audience
Vendor RFI Template	Main RFI with all requirement sections	Procurement
RFI Technical Questionnaire	Detailed questions with rationale, examples, vendors	Vendors, Evaluators
RFI Technical Questions Bank	Reference for selecting/customizing questions	Evaluators
Vendor Technical Evaluation Scorecard	Weighted scoring worksheet	Evaluators
Vendor Security Assessment	Security-focused questionnaire	Security, Vendors
RFI Response Checklist	Quick submission checklist	Vendors
Technical Compliance Matrix	Compliance form (Y/P/N/R format)	Vendors
Vendor SaaS Assessment	SaaS-specific evaluation	Evaluators
RFI Evaluation Summary	Multi-vendor comparison	Procurement

Template Selection Guide

Scenario	Recommended Templates
Standard RFI	Vendor RFI Template + RFI Technical Questionnaire
Security-Critical RFI	+ Vendor Security Assessment Questionnaire
SaaS Procurement	+ Vendor SaaS Assessment
Quick Compliance Check	Technical Compliance Matrix only
Evaluator Training	RFI Technical Questions Bank (rationale/scoring reference)

RFI Process Flow

- Preparation** — Use Questions Bank to select relevant questions
- Issue RFI** — Send Vendor RFI Template + Technical Questionnaire
- Vendor Response** — Vendors complete Compliance Matrix + Response Checklist
- Security Review** — Security team evaluates via Security Assessment
- Evaluation** — Score using Technical Evaluation Scorecard
- Comparison** — Summarize in RFI Evaluation Summary

Chapter 10

Golden Paths & Implementation Templates

Golden Paths

Opinionated, pre-configured project templates for common PBC implementation patterns. Each golden path includes scaffolding, CI/CD configuration, G-CAF contract stubs, and gateway documentation templates.

Golden Path	Language/Runtime	Includes
Node.js PBC	Node.js / TypeScript	Dockerfile, catalog-info.yaml, gateway checklist, package.json, src/index.ts
Python PBC	Python	Dockerfile, catalog-info.yaml, gateway checklist, pyproject.toml
.NET PBC	.NET / C#	Dockerfile, catalog-info.yaml, gateway checklist, .csproj
Go PBC	Go	Dockerfile, catalog-info.yaml, gateway checklist, go.mod

Each golden path includes a gateway initiation checklist that pre-fills required documentation for the G1 gateway, reducing time-to-first-gateway from weeks to days.

Implementation Templates

Template	Purpose
PBC Starter Template	Full PBC documentation template with all required sections
API Contract Template	OpenAPI 3.0+ template with G-CAF standard headers and error contracts
Gateway Initiation Checklist	Pre-filled checklist for Initiation Gateway submission